

Modelling and Predicting Photovoltaic Power Output from Environmental Conditions

Research Report

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(Dated: 6th October 2026)

This study evaluates how accurately photovoltaic AC power can be predicted from environmental measurements using a linear irradiance model, a physics-informed temperature-corrected model, and Random Forest regression. Using 14 days of January 2018 data from PVDAQ System 1433, models were trained chronologically and evaluated on a held-out four-day test set. The physics-informed model achieved the lowest mean absolute error (6.61 kW), while the irradiance-only Random Forest gave the lowest root mean square error (13.56 kW) and highest R^2 (0.888). Overall, POA irradiance dominated prediction, while adding temperature produced only modest improvement.

1. INTRODUCTION

Photovoltaic (PV) systems convert solar radiation directly into electrical energy through the photovoltaic effect. A PV cell consists of a semiconductor material in which incident photons generate electron-hole pairs, producing an electrical current when the cell is connected to an external circuit. Individual cells are connected to form modules, which can then be combined into larger arrays to provide the power required by a PV system. The electrical power produced by a system therefore depends on the amount of solar radiation incident on the modules as well as the electrical characteristics of the cells.[4]

The output of a PV system is not constant because its operating conditions vary with the environment. The most direct environmental influence is solar irradiance, since a greater incident irradiance provides more energy for conversion and generally produces a higher electrical output. The temperature of the PV module also affects its electrical performance. As module temperature increases, the voltage produced by the cells generally decreases, reducing the power generated by the system. Module temperature is influenced by both the ambient temperature and the incident solar radiation. Consequently, measurements of irradiance, ambient temperature and module temperature can provide useful information for predicting PV power output.

One approach to modelling this relationship is an empirical model, in which the relationship between measured environmental variables and electrical output is determined from observed data rather than being derived entirely from first-principles theory. An irradiance-based empirical model provides a simple baseline by relating the measured plane-of-array (POA) irradiance, (G), directly to the measured AC power, (P). In its simplest form, this relationship can be represented by

$$P = P_0 + kG$$

where P_0 is an intercept and k describes the dependence of power on irradiance. Although this model does not explicitly represent the temperature dependence of a PV module, it provides a useful measure of how much of the observed variation in power can be explained by irradiance alone.

A more physically motivated approach is to incorporate the effect of module temperature into the irradiance relationship. A temperature-corrected model can be written as

$$P = P_0 + kG[1 - \beta(T - T_{ref})]$$

where T is the module temperature, T_{ref} is a reference temperature, and β represents the temperature dependence of the system. This introduces a physical constraint into the prediction while retaining the simplicity of an empirical model. In contrast, data-driven methods such as Random Forest regression do not require the relationship between the environmental variables and power output to be specified in advance. Instead, the model learns patterns in the measured data and can account for nonlinear relationships between the input variables.

Comparing these approaches provides a way to assess the extent to which measured environmental conditions can explain variations in photovoltaic power output, while also examining whether incorporating additional physical information or using a more flexible data-driven model provides a meaningful improvement over a simple irradiance-based relationship.

2. METHODOLOGY

2.1 Data Source and Preparation

The analysis used publicly available photovoltaic system data from the PVDAQ database, provided through the Open Energy Information (OpenEI) platform. Data from System 1433, located at the National Renewable Energy Laboratory's Research Support Facility in Golden, Colorado, USA, were used. The system is a fixed, rooftop-mounted photovoltaic installation with a DC capacity of 449.28 kW[1].

A 14-day period from January 2018 was selected for the analysis. The measurements included plane-of-array (POA) irradiance, ambient temperature, module temperature and AC power. The measurements were recorded at 15-minute intervals. AC power was used as the prediction target, while POA irradiance, module temperature and ambient temperature were used as environmental predictors.

The selected measurements were combined according to their measurement timestamps to produce a single dataset containing the environmental variables and corresponding AC power measurements. Missing AC power measurements were retained as missing rather than being interpreted as zero or artificially interpolated.

2.2 Modelling Dataset

Only observations with a measured AC power value and POA irradiance greater than 20 Wm^{-2} were used for model training and testing. This irradiance threshold was applied to exclude periods of very low irradiance where the relationship between environmental conditions and system output is less representative of normal PV operation.

This filtering produced 374 observations for modelling. The observations were divided chronologically rather than randomly to preserve the temporal structure of the data. Measurements from 1–10 January were used for training, while measurements from 11–14 January were reserved as an independent test set. The training set contained 263 observations, and the test set contained 111 observations.

2.3 Outlier Screening

Potential anomalous observations were screened using the training data only. An initial linear relationship between POA irradiance and AC power was fitted to the training set, and the standard deviation of its residuals was calculated. Observations with absolute residuals greater than 1.5 standard deviations from this initial relationship were identified as potential outliers and excluded from the subsequent model fitting.

This procedure was applied only to the training data. No observations were removed from the primary test set based on their prediction errors, ensuring that the final performance evaluation remained independent of the outlier-screening procedure.

2.4 Empirical Irradiance Model

The first model provided a simple empirical baseline relating AC power to POA irradiance. A linear regression of the form

$$P = P_0 + kG$$

was fitted to the screened training data, where P is AC power, G is POA irradiance, P_0 is the fitted intercept and k is the irradiance coefficient.

This model represents the simplest relationship between the available environmental data and PV output and provides a baseline against which the more physically informed and data-driven models can be compared.

2.5 Physics-Informed Model

A second model incorporated the expected dependence of PV performance on module temperature. The model was defined as

$$P = P_0 + kG[1 - \beta(T - T_{ref})]$$

where T is module temperature, T_{ref} is a reference temperature of 25 C and β represents the temperature coefficient.

The parameters P_0 , k , and β were fitted using the screened training data. This model retains the empirical relationship between irradiance and power while introducing a physically motivated correction for deviations in module temperature from the reference temperature.

2.6 Random Forest Regression

A Random Forest regression model was used as a data-driven approach. The primary Random Forest model used POA irradiance, module temperature and ambient temperature as input features, with AC power as the target variable.

Random Forest regression combines the predictions of multiple decision trees, allowing nonlinear relationships between the environmental variables and PV output to be represented without requiring a predefined mathematical relationship. A second Random Forest model using POA irradiance alone was also evaluated to determine whether the additional temperature measurements provided meaningful predictive information.

Both Random Forest models were trained using the same chronological training set and evaluated using the same independent test set as the other models.

2.7 Model Evaluation

Model performance was evaluated using the independent test set. Three metrics were calculated: mean absolute error (MAE), root mean square error (RMSE), and the coefficient of determination R^2 .

The MAE was calculated as

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |P_i - \hat{P}_i|$$

where P_i is the measured power and $\hat{P}_i$ is the predicted power. MAE represents the average magnitude of the prediction error.

The RMSE was calculated as

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (P_i - \hat{P}_i)^2}$$

Unlike MAE, RMSE gives greater weight to large prediction errors. The coefficient of determination was also calculated to quantify the proportion of the variation in measured AC power explained by each model.

2.8 Residual and Energy Analysis

Residuals were calculated as the difference between measured and predicted AC power. The largest Random Forest residuals were examined individually and in the context of neighbouring measurements to determine whether large deviations occurred as isolated observations or persisted over several measurements. Residuals were also examined against POA irradiance and module temperature to identify possible systematic relationships.

Finally, the measured and predicted power values were integrated over time to estimate energy across valid measurement intervals. Only consecutive observations

separated by no more than 30 minutes were included in the primary energy calculation, preventing long gaps in the dataset from being interpreted as continuous operation. A sensitivity analysis was also performed using maximum allowable gaps of 15, 30 and 45 minutes to assess how the treatment of missing intervals affected the estimated energy differences.

3. RESULTS

3.1 Model Performance

The three modelling approaches produced similar overall prediction accuracy on the independent test set, although their performance differed slightly depending on the evaluation metric. The linear irradiance model achieved a mean absolute error (MAE) of 6.83 kW, a root mean square error (RMSE) of 13.81 kW, and an R^2 value of 0.884.

The physics-informed model produced the lowest MAE of 6.61 kW, indicating the smallest average absolute prediction error. Its RMSE was 13.84 kW, with an R^2 value of 0.883. Although the temperature correction reduced the average absolute error slightly compared with the linear irradiance model, it did not produce an improvement in either RMSE or R^2 .

The Random Forest model using POA irradiance, module temperature and ambient temperature produced an MAE of 6.83 kW, an RMSE of 13.58 kW, and an R^2 of 0.887. A second Random Forest model using POA irradiance alone produced an MAE of 6.89 kW, an RMSE of 13.56 kW, and an R^2 of 0.888.

Overall, the differences between the models were relatively small. The physics-informed model provided the lowest MAE, while the irradiance-only Random Forest produced the lowest RMSE and highest R^2 .

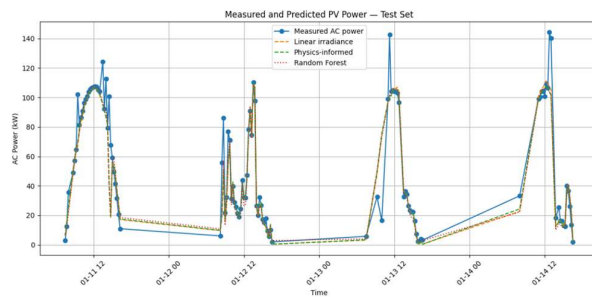


Figure 1 Measured AC power and predicted PV power for the independent test set. Predictions from the linear irradiance, physics-informed, and Random Forest models are compared with the measured AC power over the test period.

3.2 Random Forest Feature Importance

The Random Forest model using all three environmental variables showed that POA irradiance dominated the prediction. Its feature importance was approximately 0.996 for POA irradiance, compared with 0.002 for module temperature and 0.002 for ambient temperature.

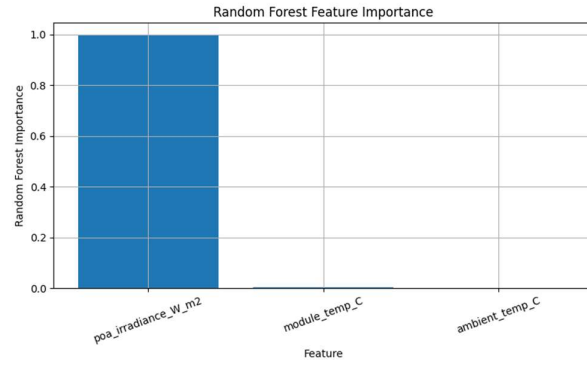


Figure 2 Shows feature importance of the input variables in the Random Forest model using POA irradiance, module temperature, and ambient temperature.

This indicates that, within the selected dataset and model, almost all of the predictive information captured by the Random Forest was associated with POA irradiance. Module and ambient temperature contributed comparatively little to the model's predictions.

3.3 Residual Analysis

The Random Forest residuals were generally distributed around zero, although several observations exhibited substantially larger errors than the majority of the test set. The standard deviation of the Random Forest residuals was 13.22 kW. Eleven test observations exceeded a sensitivity threshold of approximately 19.82 kW, corresponding to 1.5 standard deviations of the residual distribution.

These observations were retained in the primary evaluation and were examined separately rather than removed from the test set.

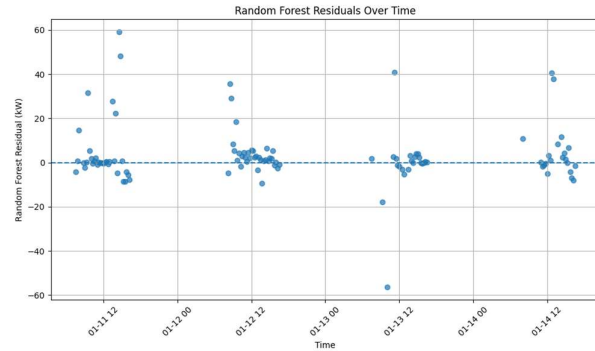


Figure 3 Random Forest residuals over the independent test period, showing the difference between measured and predicted AC power at each observation.

Figure 3 shows that the larger residuals occurred at several isolated points throughout the test period rather than forming one continuous region of large error. Examples include deviations of approximately +59.0 kW on 11 January, -56.3 kW on 13 January, and +40.6 kW on 14 January.

The residuals were also examined against POA irradiance and module temperature as shown in figure 4.

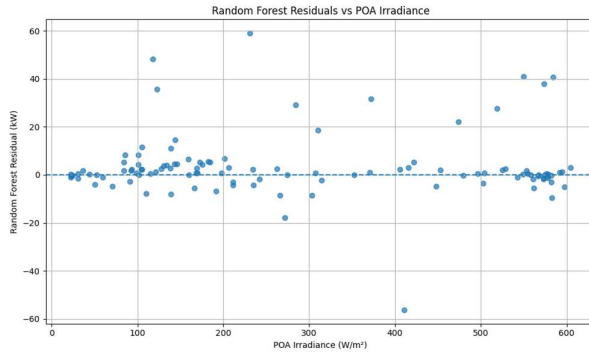


Figure 4 Random Forest residuals as a function of POA irradiance for the independent test set.

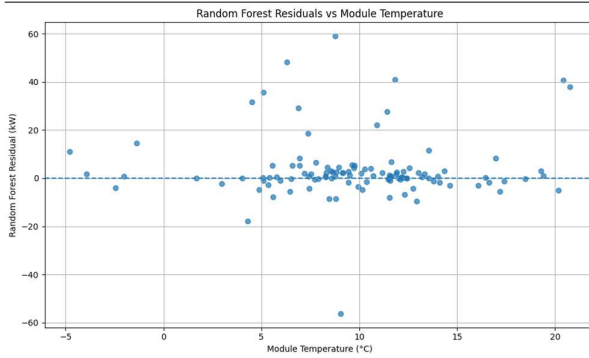


Figure 5 Random Forest residuals as a function of module temperature for the independent test set.

The residual analysis did not show a strong systematic dependence that would indicate that the remaining prediction errors were explained primarily by either irradiance or module temperature. The relatively large isolated deviations suggest that factors beyond the selected environmental variables may contribute to short-term differences between measured and predicted AC power.

3.4 Energy Comparison

The measured and predicted power values were integrated across valid consecutive measurement intervals to compare the estimated energy represented in the test dataset. Using the primary 30-minute maximum-gap criterion, the measured energy across these valid intervals was 1500.48kWh.

The linear irradiance model predicted 1392.21kWh, corresponding to a relative error of -7.22%. The physics-informed model predicted 1392.39kWh, corresponding to -7.20%, while the Random Forest predicted 1390.29kWh, corresponding to -7.34%.

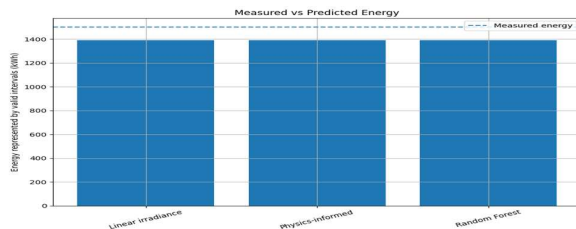


Figure 6 Measured and predicted energy across the valid measurement intervals of the independent test set.

The three models therefore produced similar energy estimates, with all three underestimating the measured energy by approximately 7% across the valid intervals represented in the test dataset. The relatively small differences between the models are consistent with their similar point-prediction performance.

3.5 Gap Sensitivity

The effect of the treatment of missing time intervals was examined by repeating the energy calculation using maximum allowable gaps of 15, 30 and 45 minutes. The relative energy errors remained within approximately 6–7% across these gap thresholds.

For the linear irradiance model, the relative error changed from approximately -7.09% using a 15-minute maximum gap to -7.22% using 30 minutes and -5.84% using 45 minutes. The corresponding values for the physics-informed model were approximately -7.03%, -7.20%, and -5.85%, while the Random Forest gave approximately -7.20%, -7.34%, and -6.15%.

This indicates that the treatment of missing intervals affects the absolute energy estimate, although the overall conclusion that the models underestimate the measured energy remains consistent.

4. ERROR ANALYSIS

The prediction errors were examined using the Random Forest model, which provided one of the strongest overall performances in the test set. The residuals were calculated as the difference between the measured and predicted AC power. Most residuals remained relatively close to zero, indicating that the model generally captured the overall variation in photovoltaic output. However, several observations showed substantially larger deviations.

The standard deviation of the Random Forest residuals was 13.22kW. Using 1.5 standard deviations as a sensitivity threshold identified 11 observations with relatively large absolute residuals. These observations were retained in the primary test-set evaluation and were examined separately to assess the nature of the prediction errors.

The largest deviations occurred at isolated points throughout the test period. For example, residuals of approximately +59.0kW and -56.3kW were observed on 11 and 13 January, respectively. The presence of these relatively large individual errors explains why the RMSE was substantially higher than the MAE. While the MAE represents the typical magnitude of prediction error, RMSE gives greater weight to these larger deviations.

The residuals were also examined as a function of POA irradiance and module temperature. No strong systematic relationship was observed that would indicate that the remaining errors were primarily associated with either variable. This is consistent with the Random Forest feature-importance results, in which POA irradiance accounted for approximately 99.6% of the model's feature importance, while module and ambient temperature each contributed approximately 0.2%.

The remaining errors therefore indicate that environmental measurements alone do not fully describe the short-term

variation in AC power in this dataset. The isolated nature of several large residuals suggests that additional system-level or temporal factors may contribute to the observed differences. However, the available data do not provide sufficient evidence to attribute these deviations to a specific physical cause.

5. DISCUSSION AND CONCLUSION

The results show that photovoltaic AC power can be predicted reasonably well from environmental measurements, with POA irradiance providing the dominant source of predictive information. All four models achieved R^2 values of approximately 0.88, indicating that a substantial proportion of the variation in measured AC power was captured by the models.

The physics-informed model produced the lowest MAE of 6.61kW, compared with 6.83kW for the linear irradiance model. This indicates that incorporating the temperature correction provided a small improvement in the average absolute prediction error. However, its RMSE of 13.84kW was slightly higher than that of the linear model, suggesting that the temperature correction did not reduce the influence of the largest prediction errors.

The Random Forest models produced similarly strong results. The model using POA irradiance, module temperature and ambient temperature achieved an RMSE of 13.58kW and an R^2 of 0.887. The POA-only Random Forest performed marginally better, with an RMSE of 13.56kW and an R^2 of 0.888. The feature-importance analysis showed that POA irradiance accounted for approximately 99.6% of the predictive importance in the full Random Forest model, while module and ambient temperature each contributed approximately 0.2%. Within this dataset, therefore, the addition of temperature measurements provided little improvement in predictive performance.

The residual analysis provides further insight into the differences between the models. Although most predictions followed the measured power reasonably closely, several isolated observations produced substantially larger errors. These errors had a noticeable effect on the RMSE while having a smaller effect on the MAE. Examination of the residuals against irradiance and module temperature did not reveal a strong systematic relationship, indicating that the remaining errors cannot be explained primarily by either of these variables.

The energy analysis produced a similar result. Across the valid measurement intervals in the test dataset, the measured energy was 1500.48kWh. The linear, physics-informed and Random Forest models underestimated this value by approximately 7.22%, 7.20% and 7.34%, respectively. The similarity of these values indicates that none of the modelling approaches provided a substantial advantage in estimating the accumulated energy over the represented intervals.

5.1 Conclusion

Overall, the results demonstrate that POA irradiance is the dominant predictor of PV power within the selected dataset. The physics-informed model achieved the lowest MAE, showing a modest benefit from explicitly incorporating

module temperature, while the POA-only Random Forest achieved the lowest RMSE and highest R^2 . However, the differences between the models were small.

The results therefore suggest that increasing model complexity does not necessarily produce a substantial improvement when the available environmental information is already dominated by irradiance. The remaining prediction errors, particularly the isolated large residuals, indicate that additional variables or system-level information may be required to explain short-term variations in AC power. Within the limitations of the 14-day dataset, the relatively simple irradiance-based approaches were therefore able to reproduce much of the observed behaviour of the PV system.

6. LIMITATIONS

Several limitations should be considered when interpreting the results. The analysis was based on a relatively short 14-day period from January 2018, with the test set covering only four days. The dataset also contained substantial missing AC power measurements, meaning that the energy analysis represented only the valid measurement intervals rather than the complete period.

The models were also restricted to the environmental variables available in the selected dataset. Although POA irradiance, module temperature and ambient temperature capture important influences on PV output, other factors may affect the measured AC power. The available data did not provide sufficient information to determine whether the larger residuals were associated with particular system-level or operational conditions.

The outlier screening was based on a 1.5-standard-deviation criterion applied to the training data. This was used as a defined screening procedure rather than as a universal definition of anomalous PV behaviour. The test data were not screened using their own residuals, and therefore the large errors identified during the test-set analysis were retained in the primary performance evaluation.

Finally, the results are specific to the selected PV system and time period. The relative performance of the empirical, physics-informed and Random Forest models may differ for other systems, seasons, climates or operating conditions. A longer dataset covering different seasons and a wider range of environmental conditions would therefore be required to assess how well these conclusions generalise.

7. REFERENCES

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